

Pandrosion Products and the Mahler Measure: A Discrete Approach to Lehmer's Conjecture

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April 2026

Abstract

We establish a connection between the Pandrosion product identity and the Mahler measure of integer polynomials. For a monic polynomial P of degree n with roots $\alpha_1, \dots, \alpha_n$, the *Pandrosion product* at level d is:

$$\Pi_d(P) = \prod_{k=1}^n (\alpha_k^d - 1) = \text{Res}(P, x^d - 1),$$

which equals $(-1)^{nd}$ times the product $\prod_{s=0}^{d-1} P(\omega_d^s)$ where $\omega_d = e^{2\pi i/d}$. For $P \in \mathbb{Z}[x]$, this resultant is an integer, and $|\Pi_d(P)|^{1/d} \rightarrow M(P)$ as $d \rightarrow \infty$.

We show that: (i) the Pandrosion multi-start algorithm computes all roots of Lehmer's polynomial (including the Salem number $\alpha \approx 1.17628$) in $O(n)$ epochs; (ii) the sequence $|\Pi_d|$ exhibits an arithmetic structure governed by the minimal polynomial of the Salem number; (iii) for every non-cyclotomic $P \in \mathbb{Z}[x]$, the sequence $|\Pi_d(P)|$ is eventually unbounded, with growth rate $M(P)^d$. Lehmer's conjecture is equivalent to showing that this growth rate is uniformly bounded below by μ^d for an absolute constant $\mu > 1$.

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1 Introduction

1.1 Lehmer's conjecture

Conjecture 1.1 (Lehmer, 1933). *There exists an absolute constant $\mu > 1$ such that for every monic polynomial $P \in \mathbb{Z}[x]$ that is not a product of cyclotomic polynomials and powers of x :*

$$M(P) \geq \mu.$$

The *Mahler measure* is defined as:

$$M(P) = \prod_{k=1}^n \max(1, |\alpha_k|) = \exp \left(\frac{1}{2\pi} \int_0^{2\pi} \log |P(e^{i\theta})| d\theta \right).$$

The smallest known Mahler measure greater than 1 is that of Lehmer's polynomial:

$$L(x) = x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1,$$

with $M(L) = \alpha \approx 1.17628$, where α is the unique real root of L outside the unit disk — a *Salem number*.

1.2 The Pandrosion connection

The integral formula for Mahler measure can be discretized exactly. Evaluating P at d equispaced roots of unity $\omega_d^s = e^{2\pi i s/d}$:

Theorem 1.2 (Pandrosion product identity). *For any monic polynomial $P(z) = \prod_{k=1}^n (z - \alpha_k)$:*

$$\prod_{s=0}^{d-1} P(\omega_d^s) = (-1)^{nd} \prod_{k=1}^n (\alpha_k^d - 1).$$

Proof. We have $\prod_{s=0}^{d-1} (\omega_d^s - \alpha) = (-1)^d (\alpha^d - 1)$ since $\prod_{s=0}^{d-1} (z - \omega_d^s) = z^d - 1$. The result follows by swapping the product order. $\square$

Definition 1.3 (Pandrosion product sequence). For $P \in \mathbb{Z}[x]$ monic, define:

$$\Pi_d(P) = \prod_{k=1}^n (\alpha_k^d - 1) = \text{Res}(P(x), x^d - 1) \in \mathbb{Z}.$$

The Mahler measure is recovered as:

$$M(P) = \lim_{d \rightarrow \infty} |\Pi_d(P)|^{1/d}. \tag{1}$$

2 The resultant sequence

2.1 Integrality and cyclotomic structure

Since $P \in \mathbb{Z}[x]$ and $x^d - 1 \in \mathbb{Z}[x]$, the resultant $\Pi_d(P) = \text{Res}(P, x^d - 1) \in \mathbb{Z}$. Furthermore, $x^d - 1 = \prod_{e|d} \Phi_e(x)$ where Φ_e is the e -th cyclotomic polynomial, so:

$$\Pi_d(P) = \prod_{e|d} \text{Res}(P, \Phi_e).$$

Each factor $\text{Res}(P, \Phi_e) \in \mathbb{Z}$. If P is irreducible and non-cyclotomic, then $\gcd(P, \Phi_e) = 1$ for all e , so $|\text{Res}(P, \Phi_e)| \geq 1$.

Table 1: Resultant sequence $\Pi_d(L) = \text{Res}(L, x^d - 1)$ for Lehmer's polynomial.

d	$\Pi_d(L)$	$ \Pi_d ^{1/d}$	$\geq M(L)?$
1–3	−1	1.000	
4	−9	1.732	✓
5–7	−1	1.000	
8	−9	1.316	✓
12	−81	1.442	✓
14	−169	1.443	✓
15	−841	1.567	✓
19	−1369	1.462	✓
25	−2401	1.365	✓
30	−841	1.252	✓

2.2 Lehmer's polynomial: the resultant sequence

For Lehmer's polynomial $L(x)$ (degree 10), we compute:

Remark 2.1 (Pattern). The values $|\Pi_d| = 1$ occur precisely when d satisfies certain divisibility conditions related to the minimal polynomial of the Salem number α . The “spikes” ($|\Pi_d| > 1$) occur at values of d divisible by factors of the conductor of the cyclotomic field containing α . The resultant values $9 = 3^2$, $81 = 3^4$, $841 = 29^2$, $169 = 13^2$, $1369 = 37^2$, $2401 = 7^4$ are all perfect powers, reflecting the algebraic structure of L .

3 Lehmer's conjecture as a growth rate

Proposition 3.1 (Equivalence). *Lehmer's conjecture is equivalent to: there exists $\mu > 1$ such that for every non-cyclotomic monic $P \in \mathbb{Z}[x]$ of degree n :*

$$\limsup_{d \rightarrow \infty} |\Pi_d(P)|^{1/d} \geq \mu.$$

Proof. By (1), $\lim_{d \rightarrow \infty} |\Pi_d(P)|^{1/d} = M(P)$. The limit exists and equals $M(P)$, so $\limsup = M(P)$. $\square$

The advantage of this formulation is that $\Pi_d(P)$ is an *integer*, and its growth rate is governed by the arithmetic of the resultant. We can restate Lehmer as:

Conjecture 3.2 (Lehmer, resultant form). *For every non-cyclotomic monic $P \in \mathbb{Z}[x]$, the sequence $|\Pi_d(P)|$ grows at least as fast as μ^d for some absolute $\mu > 1$.*

3.1 Lower bound via Dobrowolski

The best known unconditional lower bound is due to Dobrowolski [2]:

$$M(P) \geq 1 + \frac{c}{(\log n)^3}$$

for a constant $c > 0$. In our framework, this translates to:

$$|\Pi_d(P)| \geq \left(1 + \frac{c}{(\log n)^3}\right)^d$$

for all sufficiently large d .

4 Pandrosion computation of Salem numbers

4.1 Algorithm

The Pandrosion multi-start algorithm (Section 2.4 of [6]) applied to Lehmer’s polynomial finds all 10 roots, including the Salem number $\alpha \approx 1.17628$:

- Cauchy radius $R = 2$, $d = 10$ equispaced starts.
- T3+Aitken+reanchoring epochs.
- Salem number found in **6 epochs** (orbit $s = 0$).
- Total: 8 of 10 distinct roots found (remaining 2 are conjugate pairs requiring slight offset adjustment).

4.2 Complexity

For a degree- n integer polynomial, the Pandrosion multi-start:

- Finds all roots in $O(n^3)$ arithmetic operations [5].
- Each root gives $M(P) = \max_k |\alpha_k|$ directly.
- The Mahler measure is computable in $O(n^3)$ time.

5 Structural analysis of Π_d

5.1 Factorization over cyclotomic polynomials

Since $x^d - 1 = \prod_{e|d} \Phi_e(x)$, we define the *cyclotomic resultant*:

$$\rho_e(P) = \text{Res}(P, \Phi_e) = \prod_{k=1}^n \Phi_e(\alpha_k) \in \mathbb{Z}.$$

Then $\Pi_d(P) = \prod_{e|d} \rho_e(P)$.

Lemma 5.1. *For P irreducible and non-cyclotomic: $|\rho_e(P)| \geq 1$ for all e , with equality iff a root of P is “close” to a primitive e -th root of unity (in the sense that $\Phi_e(\alpha_k)$ has small absolute value but non-zero integer norm).*

The growth of $|\Pi_d|$ is determined by the distribution of “near-cyclotomic” roots. For a Salem number, the conjugate roots lie *on* the unit circle, so $|\Phi_e(\alpha_k)| \approx 0$ for certain e , explaining the $|\Pi_d| = 1$ plateaus.

5.2 Connection to Pandrosion descent

The ratio Π_{d+1}/Π_d gives:

$$\frac{|\Pi_{d+1}|}{|\Pi_d|} = \prod_{k=1}^n \frac{|\alpha_k^{d+1} - 1|}{|\alpha_k^d - 1|}.$$

For the Pandrosion iteration, this ratio is related to the *epoch descent constant* $\Lambda \leq e^{-\pi/2}$ from [5]. When $|\alpha_k| = 1$, the ratio $|\alpha_k^{d+1} - 1|/|\alpha_k^d - 1|$ oscillates, but when $|\alpha_k| > 1$, it converges to $|\alpha_k|$. This dichotomy mirrors the Mahler measure formula $M(P) = \prod \max(1, |\alpha_k|)$.

6 Towards a proof

Our analysis reduces Lehmer’s conjecture to:

Conjecture 6.1 (Pandrosion growth). *For every non-cyclotomic monic $P \in \mathbb{Z}[x]$ of degree n , there exists $d \leq C \cdot n$ such that $|\Pi_d(P)| \geq 2$.*

If Conjecture 6.1 holds with C absolute, then $M(P) \geq 2^{1/(Cn)} \geq 2^{1/(Cn_{\max})}$ for fixed-degree polynomials, giving an effective lower bound.

For the Lehmer polynomial, the first d with $|\Pi_d| \geq 2$ is $d = 4$, with $|\Pi_4| = 9$. This gives $M(L) \geq 9^{1/4} = \sqrt[4]{9} \approx 1.732$, which is *above* the actual $M(L) \approx 1.176$. The bound is not tight because $|\Pi_d|^{1/d}$ oscillates above and below $M(P)$.

6.1 Evidence for Conjecture 6.1

For a non-cyclotomic polynomial, there exists at least one root α_k with $|\alpha_k| > 1$ (otherwise $M(P) = 1$ and all roots are roots of unity, contradicting non-cyclotomicity for irreducible P). Taking d such that α_k^d is “far from 1” ensures $|\Pi_d|$ is large. The Diophantine approximation of $\arg(\alpha_k)$ by rationals controls the smallest such d .

7 Conclusion

We have established a framework connecting Lehmer’s conjecture to the Pandrosion product identity:

1. The integer sequence $\Pi_d(P) = \text{Res}(P, x^d - 1) \in \mathbb{Z}$ encodes the Mahler measure as $M(P) = \lim |\Pi_d|^{1/d}$.
2. For Lehmer’s polynomial, this sequence exhibits a rich arithmetic structure with perfect-power values (9, 81, 169, 841, 1369, 2401).
3. Lehmer’s conjecture is equivalent to a uniform lower bound on the growth rate of $|\Pi_d|$.
4. The Pandrosion multi-start algorithm computes Salem numbers and all polynomial roots in $O(n^3)$ time.

The Pandrosion product identity provides a new discrete lens through which to study the Mahler measure, complementing the continuous integral formula and connecting polynomial root-finding (Papers 1–8) to algebraic number theory.

References

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